

SCS 1307

Probability & Statistics

by

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Independent Events

- Events A and B are said to be independent if one of the events can occur without being affected by the other.
- That is, if A is independent of B, then the probability of the occurrence of A is unchanged by the information as to whether or not B has occurred.
- That is two events A and B are independent if;

$$P(A \cap B) = P(A) P(B)$$

Property 1:

If $P(A) > 0$ and $P(B) > 0$, then A and B are independent iff

$$P(A \mid B) = P(A) \quad \text{and}$$

$$P(B \mid A) = P(B)$$

Proof: If A and B are independent $P(A \cap B) = P(A) P(B)$,
then $P(A | B) = P(A)$ and $P(B | A) = P(B)$

Using conditional probability

$$P(A | B) = P(A \cap B) / P(B)$$

Substitute independence:

$$P(A | B) = P(A) P(B) / P(B)$$

$$\text{Therefore } P(A | B) = P(A)$$

Proof: $P(A|B)=P(A) \Rightarrow A$ and B are independent

Now Consider $P(A | B) = P(A)$ is true

Using conditional probability formula;

Then $P(A | B) = P(A \cap B) / P(B)$

Substitute the assumption:

$$P(A \cap B) / P(B) = P(A)$$

$$P(A \cap B) = P(A) P(B)$$

which means A and B are independent

Property 2

A and B are independent events implies that A and B' are independent events

Independence of more than two events

- We illustrate the generalisation of the concept of independence to more than two events with reference to independence of three events.

Definition

- Three events A , B and C are mutually independent if and only if
$$P(A \cap B) = P(A) P(B) ,$$
$$P(A \cap C) = P(A) P(C) ,$$
$$P(B \cap C) = P(B) P(C) \text{ and}$$
$$P(A \cap B \cap C) = P(A) P(B) P(C)$$

Property

- If A , B and C are mutually independent events, then
 - a). A and $B \cup C$ are independent events
 - b). A and $B \cap C$ are independent events
 - c). A , B and C' are mutually independent events
 - d). A , B' and C' are mutually independent eventsetc.

Examples

- A die is thrown twice. Find the probability of obtaining a 4 on the first throw and an odd number on the second throw. Check whether the A and B are independent.

Solution

Let A: “4 is obtained on the first throw”. Then $P(A) = 1/6$

Let B: “an odd number is obtained on the second die”.

Then $P(B) = 3/6 = 1/2$

The result on the second throw is not affected in any way by the results on the first throw. Therefore A and B are independent events.

And also $P(A \cap B) = 3/36$ (show!!!)

That is $P(A \cap B) = 3/36 = 1/6 * 1/2 = P(A) * P(B)$

A and B are independent.

Examples...

- A bag contains 5 red counters and 7 black counters. A counter is drawn from the bag, the colour is noted and the counter is replaced. And a second counter is drawn. Let A be the event that the first counter is red and B be the event that the second counter is black. Are the events A and B independent? Find $P(A \cap B)$.

Solution...

- Let A : “the first counter is red”. Then $P(A) = 5/12$

Let B : “the second counter is black”

since the first counter is replaced before the second draw is made, A and B are independent events.

Now, $P(B) = 7/12$

$$\text{and } P(A \cap B) = P(A)P(B) = \left(\frac{5}{12}\right)\left(\frac{7}{12}\right) = \left(\frac{35}{144}\right)$$

Therefore A and B are independent.

Examples...

- Events A and B are such that $P(A)=1/3$, $P(A \cap B)=1/12$. If A and B are independent events, find
 - a). $P(B)$ b). $P(A \cup B)$

Solution...

a). Since A and B are independent

$$P(A \cap B) = P(A) * P(B) = 1/12$$

$$\Rightarrow P(B) = 1/4$$

b).

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{3} + \frac{1}{4} - \frac{1}{12} = \frac{1}{2}$$

Examples...

- A fair die is thrown twice. Find the probability that
 - a). Neither throw results 4
 - b). At least one throw results 4

Solution...

- a) A: obtain 4 in the first throw
 B: obtain 4 in the second throw

$$P(\text{neither throw is 4}) = P(\bar{A} \cap \bar{B})$$

$$= P(\bar{A}) \times P(\bar{B}) = \left(\frac{5}{6}\right) \times \left(\frac{5}{6}\right) = \frac{25}{36}$$

b)

$$P(\text{at least one throw is 4})$$

$$= 1 - P(\text{neither result is 4}) = 1 - \frac{25}{36} = \frac{11}{36}$$

Examples...

- Two events A and B are such that $P(A)=1/4$, $P(A | B)=1/2$, $P(B | A)=2/3$
 - a). Are the events A and B independent?
 - b). Find $P(A \cap B)$
 - c). Are A and B mutually exclusive?
 - d). $P(B)$

Solution...

a). If A and B are independent, then $P(A | B) = P(A)$.

Since $P(A | B) = \frac{1}{2}$ and $P(A) = \frac{1}{4}$,

A and B are not independent.

b).

$$P(A \cap B) = P(B | A)P(A) = \left(\frac{2}{3}\right) \times \left(\frac{1}{4}\right) = \frac{1}{6}$$

Solution...

c). If A and B are mutually exclusive, then $P(A \cap B)=0$.

Since $P(A \cap B) = 1/6 \neq 0$ A and B are not mutually exclusive events.

d).

$$P(A | B)P(B) = P(B | A)P(A)$$

$$\left(\frac{1}{2}\right)P(B) = \left(\frac{2}{3}\right)\left(\frac{1}{4}\right)$$

$$\Rightarrow P(B) = \frac{1}{3}$$

Exercises

1) Toss a fair coin three times and observe the sequence of Heads and Tails. Define the following events

H_1 = Heads on first toss

H_2 = Heads on second toss

H_3 = Heads on third toss

A = all tosses the same

- a) Are the events H_1, H_2, H_3 mutually independent ?
- b) Are the events A, H_1 and H_2 mutually independent?